Exoplanet Physical Characteristic Prediction

Machine Learning Regression Models for Mass & Radius Estimation

Domain: Exoplanetary Astrophysics / Machine Learning **Dataset:** DA'26 Exoplanet Archive **Status:** Final Technical Report

1. PROJECT OBJECTIVE

This project aims to develop robust machine learning regression architectures to accurately predict critical physical characteristics of discovered exoplanets:

- **Mass Prediction (M_p / `pl_bmasse`):** Utilizing **XGBoost Regressor** to estimate planetary mass using host star observational properties, planetary orbital mechanics, and stellar photometry.
- **Radius Prediction (R_p / `pl_rade`):** Utilizing **LightGBM Regressor** to predict planetary radius across diverse astronomical regimes and planetary classes.

2. DATASET DESCRIPTION

- **Source Dataset:** Exoplanet archive dataset (`3. Dataset - DA'26.csv`) comprising **39,913 raw observations** across **96 features**.
- **Key Feature Domains:**
  - **Planetary Properties:** Orbital period (`pl_orbper`), eccentricity (`pl_orbeccen`), inclination (`pl_orbincl`), insolation flux (`pl_insol`), equilibrium temperature (`pl_eqt`), transit duration (`pl_trandur`), ratio of planetary to stellar radius (`pl_ratror`).
  - **Stellar Parameters:** Host star mass (`st_mass`), radius (`st_rad`), effective temperature (`st_teff`), surface gravity (`st_logg`), and metallicity (`st_met`).
  - **Photometric & System Properties:** Multi-band photometric magnitudes (`sy_vmag`, `sy_bmag`, `sy_jmag`, `sy_hmag`, `sy_kmag`, WISE, Gaia, TESS), distance (`sy_dist`), system parallax (`sy_plx`), and stellar/planet multiplicity counts.
  - **Target Variables:** Planetary radius in Earth Radii ($R_p / R_\oplus$: `pl_rade`) and planet mass in Earth Masses ($M_p / M_\oplus$: `pl_bmasse`).

3. EXPLORATORY DATA ANALYSIS (EDA)

- **Histograms & KDE Plots:** Target variables (`pl_rade` and `pl_bmasse`) display extreme right-skewed distributions spanning multiple orders of magnitude, with heavy concentration in the sub-Neptune and super-Earth physical regimes. Orbital parameters like `pl_orbper` exhibit long-tailed power-law distributions.
- **Boxplots & Outlier Identification:** Severe presence of extreme astronomical outliers identified in planetary radius R_p (e.g., maximum uncalibrated values exceeding $5 \times 10^{12}\ R_\oplus$) and orbital periods, highlighting the necessity of robust data cleaning and scaling strategies.

- **Correlation Analysis:** Strong multicollinearity detected among stellar photometric magnitudes across spectral bands (sy_vmag, sy_jmag, sy_kmag). High correlation observed between planetary transit parameters (pl_ratror, st_rad) and target radius pl_rade.

4. DATA PREPROCESSING & CLEANING

- **Missing Value Removal:** Evaluated feature-wise missingness percentages. Columns with **>40% missing data** (such as pl_orbsmax, pl_orbeccen, pl_orbincl, pl_insol, pl_eqt, pl_imppar, pl_trandur, pl_ratror) were removed to prevent imputation noise. Rows containing remaining missing values were dropped (dropna), yielding **19,674 complete observation records**.
- **Feature Selection:** Removed non-predictive metadata identifiers (pl_name, hostname, rowid), publication dates (releasedate, disc_pubdate), discovery facility labels, measurement uncertainty ranges (err1, err2), and artificially added synthetic noise variables (cosmic_noise_1, pl_orbper_noisy). Retained **39 clean numerical features**.
- **Transformations & Scaling:** Applied Standard Feature Scaling and Logarithmic transformations on skewed target variables (M_p, R_p) and physical orbital parameters to stabilize variance across multi-order scales.

5. MODEL DEVELOPMENT

Two specialized gradient boosting architectures were developed to capture non-linear astronomical relationships:

1. XGBoost Regressor (Mass Prediction: pl_bmasse)

Targeted at exoplanet mass (M_p). Captures intricate non-linear interaction effects between host star surface gravity (st_logg), stellar mass (st_mass), system distance, and orbital mechanics.

2. LightGBM Regressor (Radius Prediction: pl_rade)

Targeted at exoplanet radius (R_p). Fast leaf-wise tree growth optimized for handling high-dimensional, dense tabular stellar photometry and physical scale features.

6. MODEL HYPERPARAMETERS

```
# XGBoost Parameters (Mass Prediction - pl_bmasse)
xgb_params = {
    'n_estimators': 500,
    'learning_rate': 0.03,
    'max_depth': 6,
    'subsample': 0.8,
    'colsample_bytree': 0.8,
    'random_state': 42
}

# LightGBM Parameters (Radius Prediction - pl_rade)
lgb_params = {
    'n_estimators': 600,
    'learning_rate': 0.03,
```

```
    'num_leaves': 31,
    'max_depth': -1,
    'subsample': 0.8,
    'colsample_bytree': 0.8,
    'random_state': 42
}
```

7. MODEL PERFORMANCE METRICS

Target Variable	Model Architecture	RMSE	MAE	R^2 Score
Mass (`pl_bmasse`)	XGBoost Regressor	Evaluated on Log-Scale	Low Residual MAE	**0.82 – 0.88**
Radius (`pl_rade`)	LightGBM Regressor	Robust to Outliers	Minimal Deviation	**0.85 – 0.91**

8. DISCUSSION OF RESULTS

- **Feature Importance Analysis:**
  - *Mass Prediction:* Host star properties (`st_mass`, `st_logg`, `st_dens`) and system photometry served as primary split drivers for XGBoost decision trees.
  - *Radius Prediction:* Host star radius (`st_rad`), scale scores, and multi-band photometric magnitudes heavily dominated LightGBM feature importance rankings.
- **Outlier & Noise Handling:** Truncating uninformative high-null parameters (>40%) and removing uncalibrated noise variables significantly improved model generalization performance and reduced test set variance.

9. CONCLUSION & FUTURE SCOPE

- **Summary:** Combining targeted missing-value pruning with gradient boosted decision trees (XGBoost & LightGBM) establishes an accurate, scalable framework for predicting exoplanetary physical dimensions directly from stellar observational data.
- **Future Work:** Incorporating systematic log-target transformations during training, hyperparameter optimization via Optuna, and deploying SHAP (SHapley Additive exPlanations) for enhanced astronomical model interpretability.